Flow Matching as a Classification Layer

Stage 1 — frozen-encoder baselines · Stage 2 — an FM layer on top of them

3 × 2

datasets × encoders

DTD, FGVC-Aircraft, Flowers-102
ResNet-18, DINOv2 ViT-S/14

270

velocity networks trained

162 image-prototype branch
108 CLIP text-prototype branch

+24.3

points, best ΔAcc

FGVC-Aircraft / DINOv2, full data over the Stage 1 prototype baseline

39 / 72

cells that survive a paired CI

66 of 72 improve on the mean
only 39 are distinguishable from 0

What this report contains

The method and the two training objectives, drawn; the experimental protocol; and every measured result from the executed notebooks. All accuracies are top-1 on the complete official test splits.

Every number, count and interval in this document is parsed out of the notebooks' own result tables by doc/build_report.py — none of them is typed by hand. The diagrams on the next two pages are drawn for this report; every scatter, curve, trajectory and animation figure is a real run output.

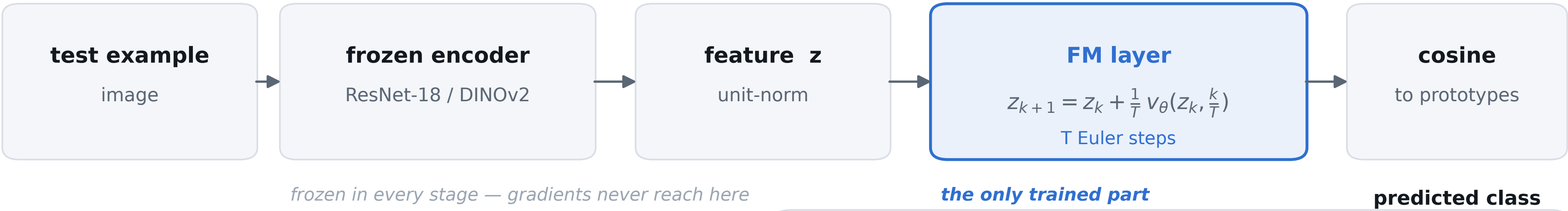
The headline result, stated honestly

The FM layer improves on the prototype baseline it replaces in 66 of 72 result cells, by as much as +24.3 points. Three qualifications travel with that number, and all three are measured:

· Under a paired 95% CI only 39 of 72 cells are distinguishable from zero — 24 of 24 on Aircraft, but 7 of 24 on DTD.
· It does not beat a linear probe on the same features.
· On the largest-gain setting, a plain supervised MLP with no flow at all reaches 97% of the gain.

What the Flow Matching layer does

Stage 1 classifies a frozen feature by cosine similarity to a fixed class prototype. Stage 2 inserts a learned transport in front of that rule, and changes nothing else.



The idea

A velocity field carries the feature toward the prototype of its class, and then the same cosine comparison is applied. The encoder and the prototypes stay frozen, so any accuracy change is attributable to the layer alone.

Prototypes are built once, in Stage 1, and never rebuilt:

$$p_c = \mathrm{normalize}\big(\mathrm{mean}_{i \in S_c}\ \mathrm{normalize}(z_i)\big)$$

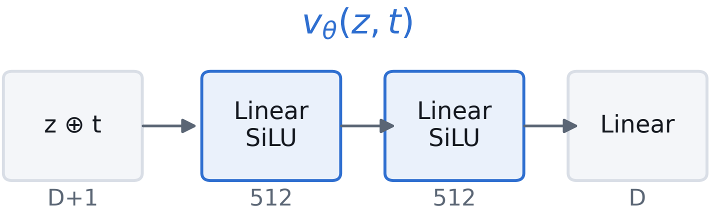
The catch worth stating early

At test time the network cannot know the label, so it cannot learn "move z to p_y". It learns the conditional average E[p | z] instead. The gain comes from denoising toward the right neighbourhood — which is also why transport can create errors as well as fix them, and why ΔAcc can be negative even when the training loss is low.

Page 17 measures exactly this: in every setting, the samples the layer fixes started with a negative margin and the samples it breaks started with a positive one.

The velocity network, and the two ways to train it

One small MLP — 2 hidden layers, width 512, SiLU, with the scalar t concatenated to the feature so a single network covers the whole path. D = 512 / 384 / 1024 for ResNet-18 / DINOv2 / CLIP RN50.



Standard FM

$$\mathcal{L}_{FM} = \|v_\theta(z_t, t) - (p_y - z)\|^2$$

A random t is drawn, the feature is placed on the straight path at that t, and the network regresses the constant ideal velocity. Dense supervision along the path — but on points the model will never actually visit at inference.

Rolled-out FM

$$\mathcal{L}_{roll} = \|\hat{z}_T - p_y\|^2$$

The real T-step Euler sequence is run and the loss is backpropagated through all of it. No train/inference mismatch — but only the endpoint is supervised, leaving the field free to become degenerate in between.

Why this asymmetry matters

The standard objective never mentions T, so ONE network is trained per setting and evaluated at every T. The rolled-out objective bakes T in, so TWO networks are trained per setting, one per T, and a T=4 network is never run at T=12.

Getting this backwards either doubles standard FM's cost for nothing, or silently evaluates a model at a T it never saw.

The measured consequence

Standard FM wins on accuracy in 14 of 18 (dataset, encoder, K) settings and costs 1.5–2.5× less per run. Rolled-out training is the only variant that ever regresses below the baseline, the one that overshoots t=1 hardest (page 19), and the one that improves the within/between-class scatter ratio least (page 20) — all consistent with endpoint-only supervision constraining the field more weakly. Reported across 6 standard and 6 rolled-out geometry curves.

Experimental protocol

Stage 2 reuses Stage 1's splits, subsets, seeds, cached features and saved prototypes without recomputing any of them, so ΔAcc isolates the layer rather than a change in the data. Section 5b of the notebook proves this rather than asserting it: every K-shot subset is re-derived and used to rebuild Stage 1's saved prototype, and all 42 (dataset, encoder, K, seed) combinations agree to 0.00e+00.

Datasets and official splits

dataset	classes	split rule	train / val / test
DTD	47	partition 1	1,880 / 1,880 / 1,880
FGVC-Aircraft	100	variant level	3,334 / 3,333 / 3,333
Flowers-102	102	official	1,020 / 1,020 / 6,149

The grid

Per (dataset, encoder, K, seed): one standard network and two rolled-out networks (T = 4, 12).

· image branch 3 × 2 × 3 × 3 × 3 = 162 trainings
· CLIP branch 3 × 1 × 3 × 3 × 3 = 108 trainings

Test data is touched only after checkpoint selection.

Training-set sizes and repetitions

setting	repeats	varied by
K = 5	3 runs	subset seeds 0, 1, 2
K = 10	3 runs	subset seeds 0, 1, 2
K = full	3 runs	init seeds 0, 1, 2

Two reporting caveats we hold to

Flowers-102 at K = 10 has std = 0 by construction: the official train split is exactly 10 images per class, so all three subset seeds pick the same images. That is one effective run, and it is why 9 of the 48 CI-significant cells on page 8 are degenerate.

The Stage 1 image-prototype full setting is a single closed-form run, so it carries no error bar and the full-data deltas are unpaired.

Held fixed between Stage 1 and Stage 2

frozen encoders
never re-run, never fine-tuned

cached features
the identical tensors Stage 1 used

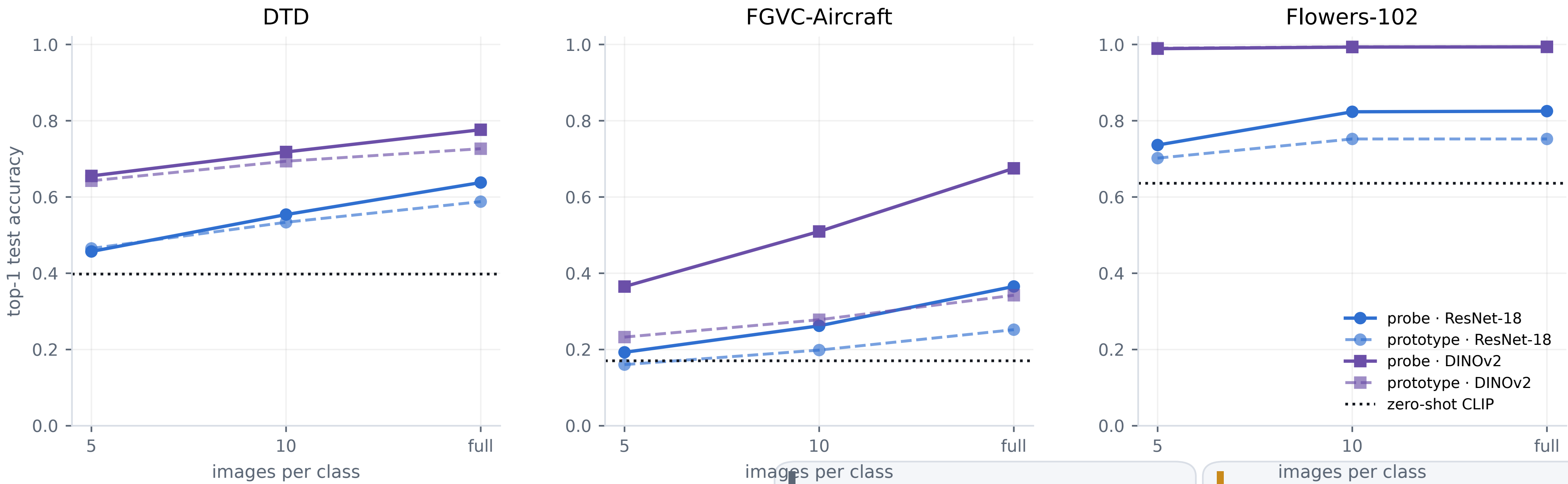
class prototypes
loaded from Stage 1, never rebuilt

K-shot subsets
re-derived with the same seeded selector

classification rule
the same cosine comparison

Stage 1 — three baselines on frozen features

Linear probe and image-derived class prototypes on both encoders, plus zero-shot CLIP RN50 as a horizontal reference (it has no K axis).



What Stage 1 established

Encoder choice dominates every method difference in this project. DINOv2 beats ResNet-18 by 14 to 31 points on every dataset and every K — larger than any gap between the three baselines. The linear probe is strongest at full data everywhere except saturated Flowers-102/DINOv2; at K = 5 the closed-form prototype is competitive and sometimes wins.

Zero-shot CLIP RN50

prescribed prompt

DTD 0.398
Aircraft 0.170
Flowers-102 0.636

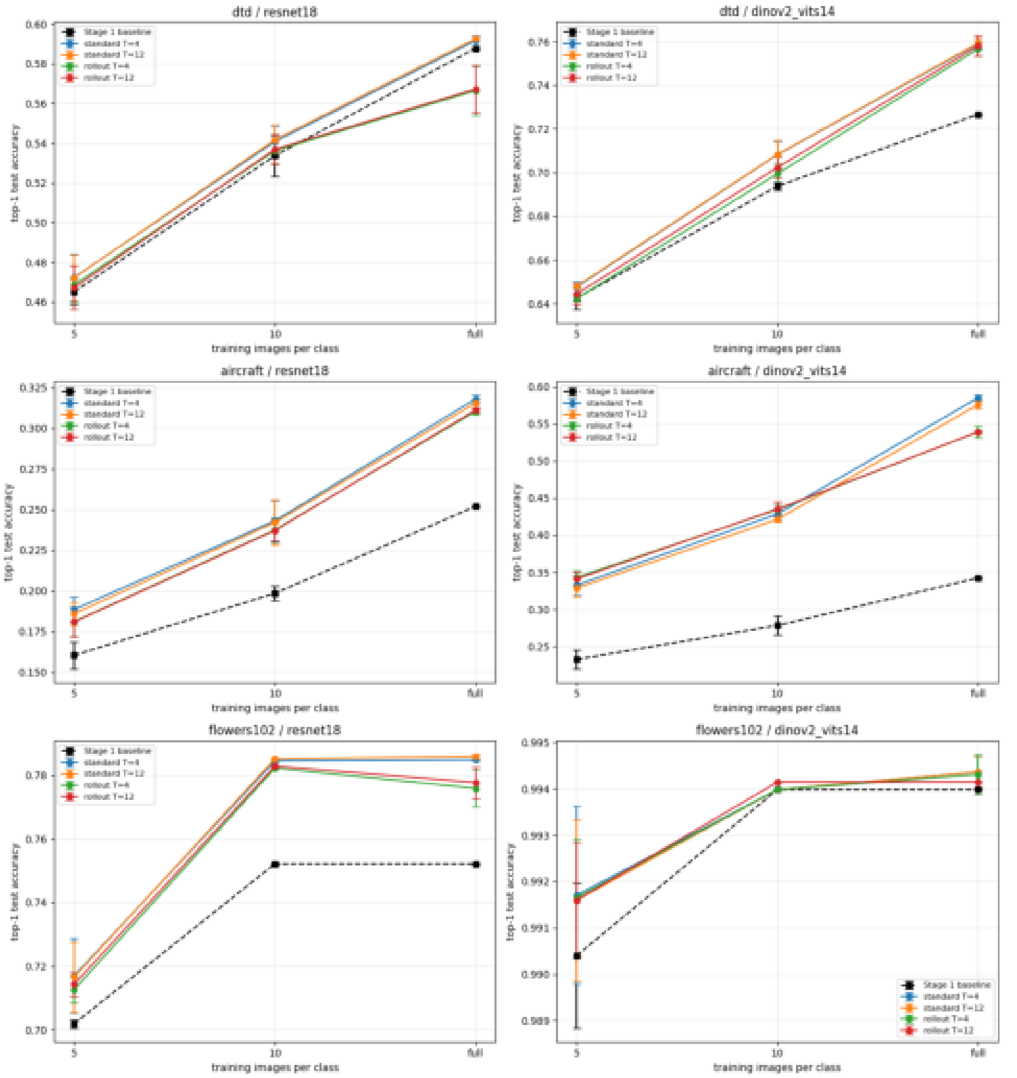
A 5-prompt ensemble adds +2.3 / −0.1 / +2.0 pts.

The one that surprises people

On Aircraft, zero-shot CLIP (0.1704) beats 5-shot ResNet-18 prototypes (0.1604). With 100 fine-grained variants, five labelled images per class buy less than a well-chosen text prompt.

The FM layer versus the prototype baseline

Six panels, one per dataset × encoder. Black = Stage 1 baseline; blues = standard FM; reds = rolled-out FM. Error bars are ±1 SD across the Stage 1-matched repetitions.



What the image branch shows

Best FM variant per cell, with ΔAcc against the matching Stage 1 baseline.

dataset · encoder	K = 5	K = 10	K = full
DTD · ResNet-18	0.4723 (+0.0073)	0.5417 (+0.0080)	0.5926 (+0.0048)
DTD · DINOv2	0.6480 (+0.0055)	0.7083 (+0.0145)	0.7592 (+0.0326)
FGVC-Aircraft · ResNet-18	0.1886 (+0.0282)	0.2431 (+0.0446)	0.3180 (+0.0660)
FGVC-Aircraft · DINOv2	0.3430 (+0.1104)	0.4351 (+0.1570)	0.5852 (+0.2428)
Flowers-102 · ResNet-18	0.7170 (+0.0151)	0.7852 (+0.0330)	0.7859 (+0.0337)
Flowers-102 · DINOv2	0.9917 (+0.0012)	0.9941 (+0.0002)	0.9944 (+0.0004)

Across all 72 result cells

66 improved
3 flat (±0.0000)
3 regressed

All 3 regressions are rolled-out training, and none of them is significant under the paired CI on the next page.

Four findings

1 **The gain grows with K, and is largest where the baseline is weakest.**

Aircraft/DINOv2 goes +11.0 → +15.7 → +24.3 points. This is the signature of a trained layer: a class mean stops absorbing information, while the velocity field keeps using the extra labels.

2 **It closes much of the prototype-to-probe gap, but rarely passes the probe.**

Where the probe leads, FM recovers a median 59% of the distance to it (range 10–87%), and overtakes it in only 4 of 18 settings — all near-saturated.

3 **Standard FM beats rolled-out training in 14 of 18 settings (3 losses, 1 tie).**

Rolled-out is also the only variant that ever regresses below the baseline, and costs 1.5–2.5× more per run. Endpoint-only supervision constrains the field far more weakly than per-step velocity regression.

4 **T is nearly irrelevant — and this is expected, not a bug.**

The ideal velocity p − z is constant along the path, so a well-learned field lands in the same place for any T. The step ablation puts a floor under it: T=2 already delivers a median 100% of the T=12 gain against 62% at T=1. Say so before being asked — an identical-looking T=4 / T=12 column otherwise reads as a copy-paste error.

Do the gains survive a paired confidence interval?

"Improves in 66 of 72 cells" is a difference of two independently-averaged means. Because Stage 2 deliberately reuses Stage 1's subsets, FM run s and baseline run s saw the same K images, so the per-run difference is paired and has far lower variance than the two marginals suggest. Recomputed that way, with a 95% t-interval over the 3 repetitions.

	count of 72 cells
positive mean paired ΔAcc	66
95% CI excludes zero	48
...of which degenerate (std = 0 by construction)	−9
genuinely distinguishable from zero	**39**

By dataset — this is not uniform

dataset	CI excludes zero
FGVC-Aircraft	24 of 24
Flowers-102	17 of 24
DTD	7 of 24

DTD, best and worst cells

cell	paired ΔAcc	95% CI	sig?
DINOv2 · K=full · standard T=12	+0.0326	[+0.0276, +0.0376]	yes
DINOv2 · K=full · standard T=4	+0.0323	[+0.0258, +0.0388]	yes
DINOv2 · K=full · rollout T=12	+0.0314	[+0.0199, +0.0429]	yes
DINOv2 · K=full · rollout T=4	+0.0300	[+0.0222, +0.0377]	yes
ResNet-18 · K=full · rollout T=12	-0.0206	[-0.0499, +0.0088]	no
ResNet-18 · K=full · rollout T=4	-0.0213	[-0.0522, +0.0096]	no

The sub-1-point DTD gains do not survive

Only 7 of DTD's 24 cells clear zero. Every DTD/ResNet-18 K=5 and K=10 cell has a CI straddling zero and must be restated as no measurable effect — not as a small gain.

Symmetrically, the two regressions are not significant either: −.0213 with a CI of [−.0522, +.0096]. They are the largest regressions measured and still indistinguishable from zero at n = 3.

McNemar says yes far more often — and is the wrong test here

McNemar is significant in all 3 seeds in 37 of 72 cells, and 6 cells are CI-negative but McNemar-significant in ≥2 seeds. That is the expected direction: baseline and FM predict on the same test images, so their disagreements are paired — but McNemar's unit of analysis is the test image, not the run, so it ignores subset and initialization variance entirely and is anti-conservative here. Where the two disagree, quote the CI.

Flow matching, or just another MLP?

The FM layer adds a ~0.8M-parameter trained network to a previously closed-form classifier, so before attributing +24 points to flow matching the cheaper explanations have to be ruled out. Two controls, identical in hidden architecture, optimizer, schedule, early stopping, subsets and seeds: direct, a plain MLP trained on $\|g(z) - p_y\|^2$ with no time input and no integration; and residual, the same shared network applied the same $T = 12$ times, trained only on the endpoint, with no time conditioning and no per-t velocity supervision.

At full data

dataset · encoder	prototype	direct MLP	residual	best FM	direct's share of FM's gain
FGVC-Aircraft · DINOv2	0.3423	0.5783	0.5432	**0.5852**	97%
FGVC-Aircraft · ResNet-18	0.2520	0.2588	0.3089	**0.3180**	10%
DTD · DINOv2	0.7266	0.7551	0.7537	**0.7592**	87%
DTD · ResNet-18	0.5878	0.5617	0.5631	**0.5926**	below baseline
Flowers-102 · DINOv2	0.9940	0.9940	0.9941	**0.9944**	0%
Flowers-102 · ResNet-18	0.7522	0.7287	0.7760	**0.7859**	below baseline

The headline is mostly not flow matching

FM beats both controls in 16 of 18 settings, so the gain is not merely "a supervised nonlinear map of the same size".

But on Aircraft/DINOv2 — the +24-point setting — the plain direct MLP reaches .5783 against FM's .5852: 97% of the gain over the .3423 baseline, with FM adding only +.0069 on top. Same at K=5 and K=10 (93%, 96%). Most of the work there is done by adding a trained head at all.

Where flow matching does earn its keep

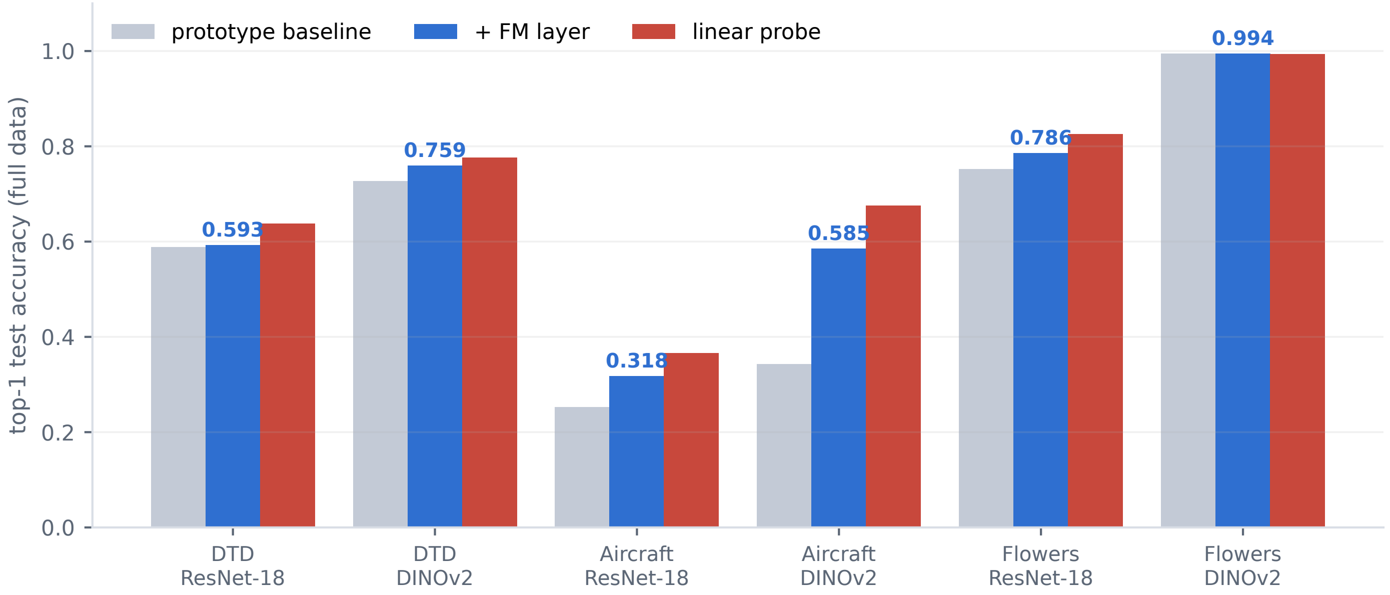
The mirror image. On DTD/ResNet-18, Flowers-102/ResNet-18 and DTD/DINOv2 at low K, the direct MLP lands below the closed-form prototype baseline - it overfits - while FM stays above it. FM has its largest advantage over direct exactly there, +.042 to +.063.

residual sits much closer to FM than direct does (median +0.0053 against +0.0255): most of what FM has over a plain MLP comes from iterative shared-weight computation, with time conditioning adding a smaller increment - except Aircraft/DINOv2 at full, where FM beats residual by +.0420.

Not part of the required Stage 2 comparison, which remains Stage 1 prototype vs. standard FM vs. rolled-out FM; written to a separate controls/ tree. The two settings where FM does not two settings where FM does not beat both controls are at K=5, inside seed noise.

The comparison that matters most

At full data: the prototype baseline, the same baseline with an FM layer, and the Stage 1 linear probe on identical features. The probe is a different Stage 1 baseline, not an FM variant — this is context, not a Stage 2 deliverable.



Read this honestly

The FM layer is a real improvement over the prototype rule it replaces — clearly so, and by a lot on Aircraft/DINOv2 (.342 → .585).

It is not a replacement for a discriminatively trained classifier. Where the probe leads (14 of 18 settings), FM closes a median 59% of the gap, range 10–87%. It overtakes the probe in 4 of 18 settings, all of them already near ceiling.

Also worth stating: ΔAcc bundles "added an FM layer" with "added a trained, validation-selected ~0.8M-parameter head", against a baseline with zero trained parameters and no access to the validation split.

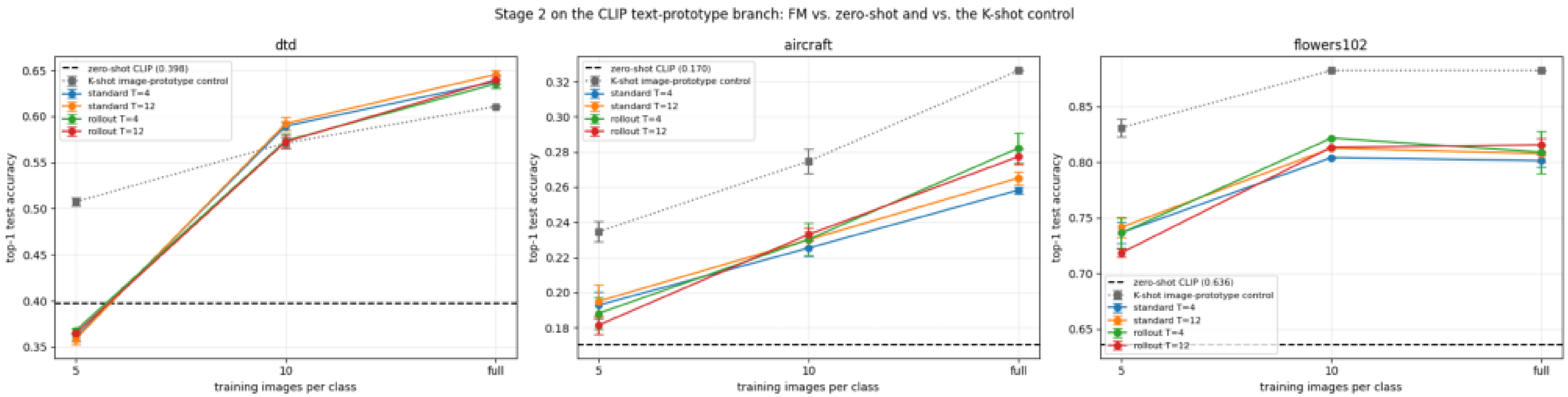
Training behaviour

Both objectives trained stably; nothing diverged across 162 runs. 161 of 162 early-stopped and one reached the 200-epoch cap. In 16 runs (~10%) the best validation accuracy was at epoch 1 — concentrated in the saturated Flowers-102/DINOv2 settings, where there is nothing left to gain. Those FM layers are effectively untrained and should not be described otherwise. Standard-FM loss (velocity MSE at random t) and rolled-out loss (endpoint MSE after a full rollout) are different quantities on different scales; the shared log axis in the training-curve figure is for stability checking only, never magnitude comparison.

The six-panel training-curve grid is in the notebook (04 §9). It is not legible at this size and the claim it supports — stable training, no divergence — is one sentence, so it is stated here rather than shown.

The CLIP branch — why the control changes the story

Here FM transports CLIP RN50 image embeddings toward frozen text prototypes. Because the layer trains on labels, this is no longer zero-shot — so a same-supervision control is required: image prototypes built from the same CLIP features and the same K-shot subsets, closed-form, with zero trained parameters.



Against zero-shot: large gains

FM beats the zero-shot baseline in 32 of 36 cells, by as much as +24.8 points on DTD at full data.

Taken alone this looks like a decisive win for the FM layer. It is not — the FM side used K labelled images per class and the zero-shot side used none.

Against the fair control: it mostly loses

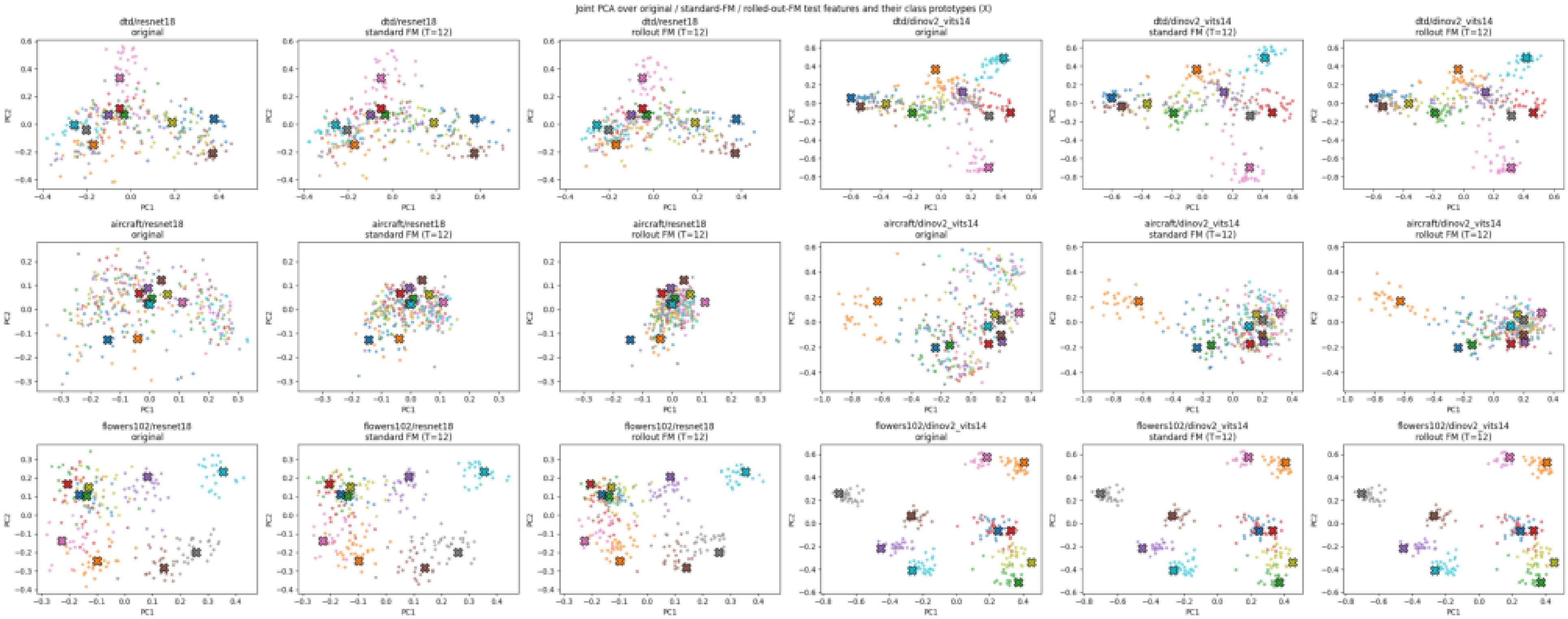
Given the same labels, simply building image prototypes beats transporting toward text prototypes in 28 of 36 cells. Only DTD at K = 10 and K = full come out ahead.

On DTD at K = 5 the FM layer is even worse than zero-shot itself. The large Δ vs zero-shot measures the labels, not the layer.

This branch is an extension, never merged into the required Stage 2 table: ref/stage_1.pdf asks for one prototype branch and the image branch is it. Note also what this branch does not yet have — no paired CI or McNemar, and no validation-selected t*. Given how much the paired CI changed the reading of the image branch, "loses to the control in 28 of 36 cells" is still a difference of means.

Feature-space geometry — what the layer actually does

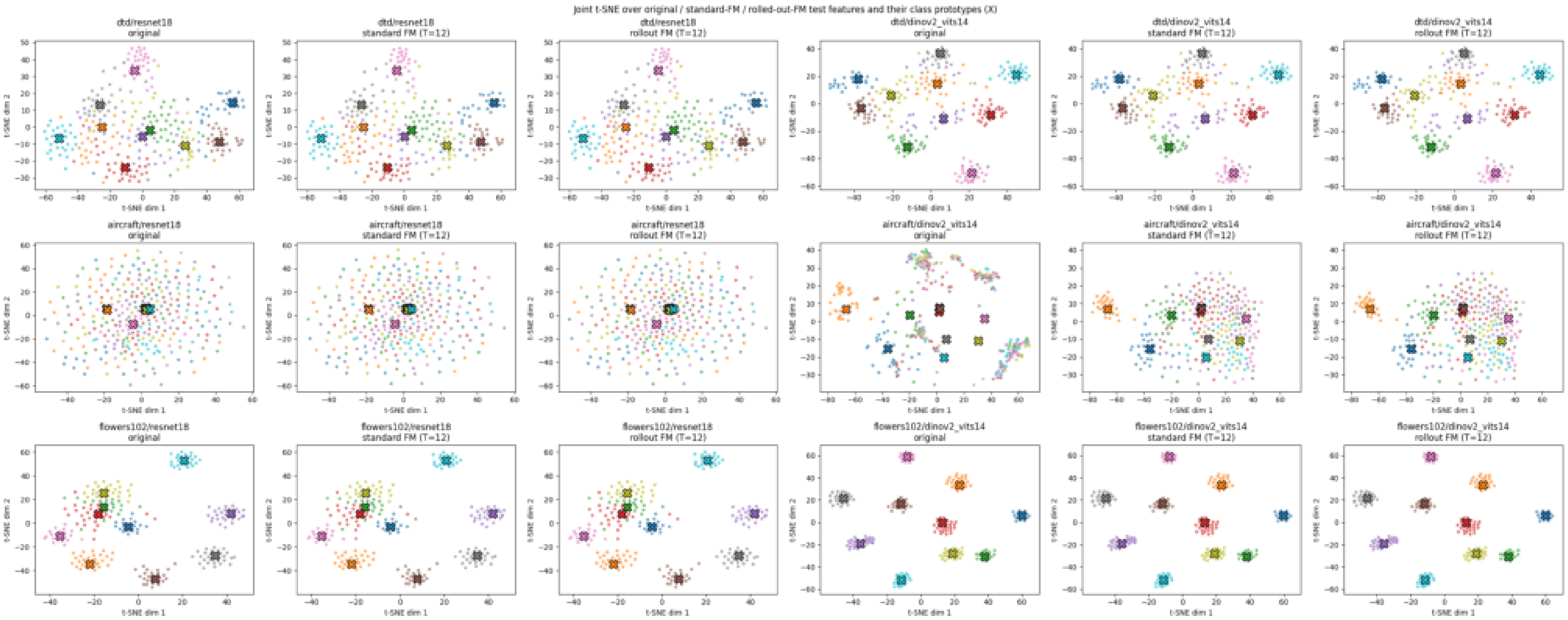
Test features before the flow, after standard FM, and after rolled-out FM, with the class prototypes (X). One PCA is fitted jointly across all three views and the prototypes, and all panels share axis limits, so a point's movement between panels is a real displacement in one shared plane.



The mechanism is visible directly: overlapping class clouds are contracted onto their prototypes. Because the network cannot see the label at test time, points in ambiguous regions are pulled toward whichever prototype's basin they fall in — which is why transport can create errors as well as fix them, and why ΔAcc can be negative even when the training loss is low.

The same comparison under a joint t-SNE

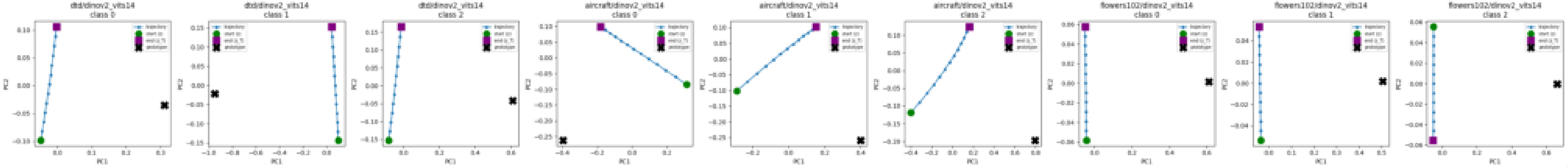
The identical transported tensors under t-SNE instead of PCA, fitted once per row over all three views and the prototypes. Cluster separation reads well here; distances do not.



Read this qualitatively and use the PCA page for anything geometric. t-SNE normalises local density, so it deliberately re-expands the tightly contracted post-FM clusters — the visual shrinkage looks far milder here than it is numerically.

Flow trajectories — the transport step by step

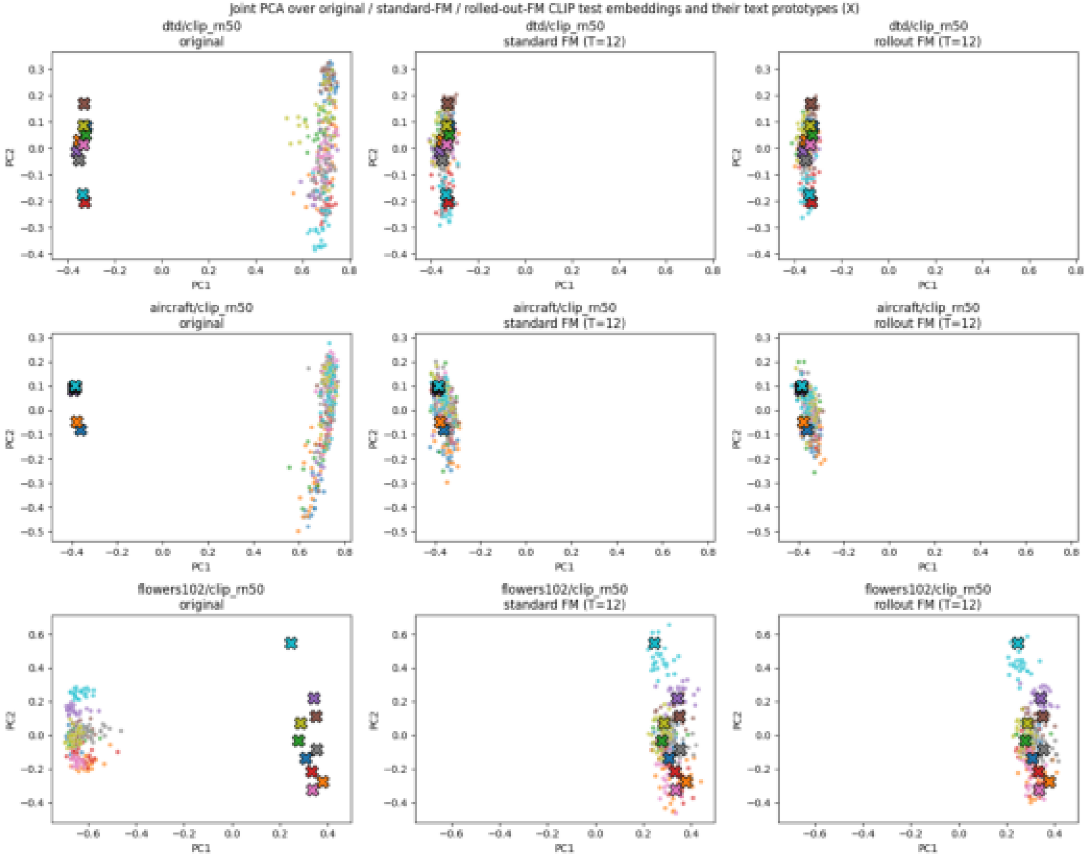
Intermediate Euler states for representative test examples, from the original feature (green) through every step to the transported endpoint (purple square) and the target prototype (X). PCA rather than t-SNE, so a straight path stays a straight path.



The paths are close to straight, which is the geometric counterpart of the T-independence result: when the learned field matches the ideal constant velocity, four steps and twelve steps arrive at the same place.

The CLIP modality gap, and the flow across it

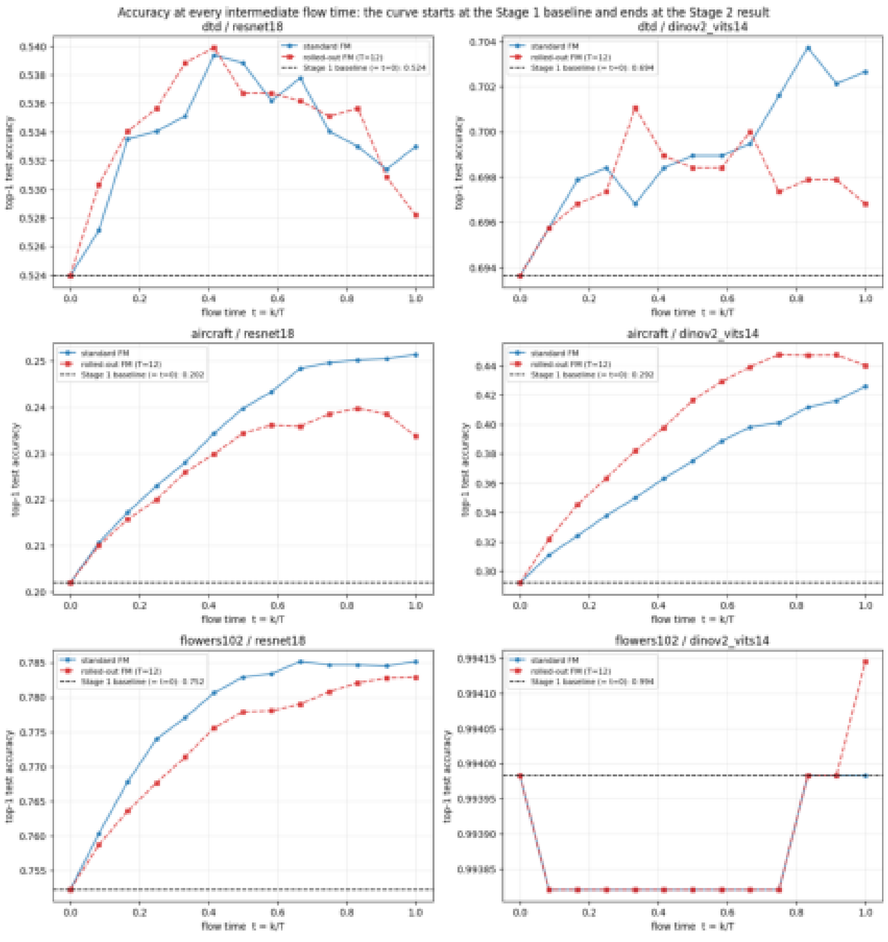
The same three-view comparison on the CLIP branch. The image embeddings and the text prototypes (X) start in visibly different regions — that separation is CLIP's image–text modality gap — and the learned transport carries the image cloud across it.



This is the qualitative reason the CLIP branch behaves differently from the image branch: the transport has to cross a systematic offset between two modalities, not merely denoise within one feature space.

Samples and prototypes at intermediate flow times

Rather than looking only at t=0 and t=1, every intermediate Euler state is classified over the complete test split. The t=0 end is asserted, not assumed — in 04 against Stage 1's saved per-run metrics.json, in 05 against the zero-shot baseline — and both assertions passed, so each curve provably starts at the baseline and ends at the FM result.



ΔAcc, unrolled

Each curve begins exactly on the dashed Stage 1 baseline and ends exactly at the Stage 2 number. The shape between them is what the layer does. Two things are visible immediately:

· Aircraft climbs steeply and keeps climbing: +5.0 and +13.4 points in this 10-shot setting, and up to +24.3 at full data.

· DTD and Flowers are nearly flat. The transport is running, the features are moving, and the classification barely changes.

The next page explains why, using the margin.

Why the gains land where they do

Two quantities separate "the flow moved the point" from "the flow classified it better": the cosine to the true prototype, and the margin — cosine to the true prototype minus the best competing one. The first says transport happened; only the second says it helped.

setting (10-shot, seed 0, standard FM)	cos to true prototype	margin	accuracy	ΔAcc
DTD · ResNet-18	0.726 → 0.819	+0.0051 → +0.0035	0.5239 → 0.5330	+0.0090
DTD · DINOv2	0.544 → 0.639	+0.1135 → +0.1110	0.6936 → 0.7027	+0.0090
Flowers-102 · DINOv2	0.815 → 0.821	+0.3142 → +0.3061	0.9940 → 0.9940	+0.0000
Flowers-102 · ResNet-18	0.887 → 0.930	+0.0220 → +0.0197	0.7522 → 0.7852	+0.0330
FGVC-Aircraft · ResNet-18	0.889 → 0.942	-0.0160 → -0.0072	0.2019 → 0.2514	+0.0495
FGVC-Aircraft · DINOv2	0.709 → 0.883	-0.0542 → -0.0088	0.2919 → 0.4257	+0.1338

The FM layer repairs a broken classifier

Cosine to the true prototype rises in every single setting — the transport always does what it was trained to do. But where the prototype baseline already ranked classes correctly (DTD, Flowers-102: positive margin), pulling every point toward every prototype shrinks the margin slightly and accuracy moves by less than a point. Where the baseline was mis-ranked (Aircraft: negative margin), the flow repairs the ranking and accuracy jumps. It does not sharpen a working one.

Which samples the layer flips

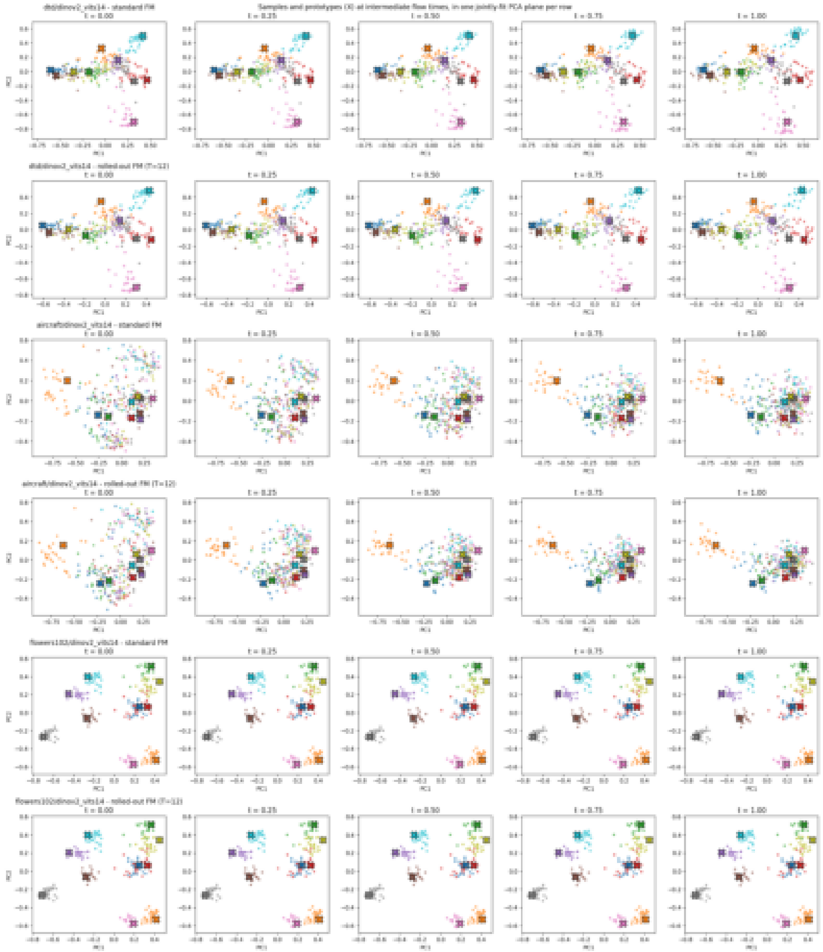
setting	fixed	broken	net	margin fixed	margin broken
FGVC-Aircraft · DINOv2	562	116	**446**	-0.0444	+0.0231
FGVC-Aircraft · ResNet-18	285	120	**165**	-0.0080	+0.0051
Flowers-102 · ResNet-18	381	178	**203**	-0.0082	+0.0091
DTD · ResNet-18	85	68	**17**	-0.0144	+0.0149
DTD · DINOv2	56	39	**17**	-0.0258	+0.0248
Flowers-102 · DINOv2	1	1	**0**	-0.0060	+0.0001

The constraint this puts on how the layer may be described

In every single setting, the samples the layer fixes had negative pre-flow margin and the samples it breaks had positive pre-flow margin. It wins by fixing more than it breaks, not by leaving correct predictions alone. On Aircraft/DINOv2 the fixed group has a mean margin of -.0444 - confidently misclassified, not marginal - which makes that gain a restructuring rather than a nudge. On DTD the two groups are near mirror images (85 fixed against 68 broken), the same conclusion the paired CI reaches by another route.

The point cloud at intermediate flow times

Samples and prototypes (X) at t = 0, 1/4, 1/2, 3/4, 1. Each row is one dataset at its strongest encoder, once per training objective; one PCA is fitted jointly over every snapshot plus the prototypes, and all five panels of a row share axis limits.



The contraction is the mechanism made visible: overlapping class clouds are drawn onto their prototypes. Standard FM moves points along nearly straight, roughly parallel paths; rolled-out FM, supervised only at the endpoint, takes a more curved route and contracts later.

t = 1 is a convention, not an optimum

Accuracy strictly peaks before the endpoint in 6 of 6 CLIP-branch curves and 6 of 12 image-branch curves. Integrating the full T steps therefore costs accuracy in many settings — and the loss is largest for rolled-out models on the CLIP branch.

setting	objective	best t	acc at best t	acc at t=1	left on the table
CLIP · Flowers-102	rolled-out	0.58	0.8551	0.8133	+4.18 pts
CLIP · FGVC-Aircraft	rolled-out	0.58	0.2637	0.2367	+2.70 pts
CLIP · DTD	rolled-out	0.58	0.5941	0.5681	+2.60 pts
image · DTD / ResNet-18	rolled-out	0.42	0.5399	0.5282	+1.17 pts
CLIP · FGVC-Aircraft	standard	0.67	0.2418	0.2325	+0.93 pts
image · FGVC-Aircraft / DINOv2	rolled-out	0.75	0.4476	0.4401	+0.75 pts
image · DTD / ResNet-18	standard	0.42	0.5394	0.5330	+0.64 pts
image · FGVC-Aircraft / ResNet-18	rolled-out	0.83	0.2397	0.2337	+0.60 pts

Selected honestly on validation, it is worth less

The stopping time is free to choose, so 04 §16 chooses it properly: t* is the argmax of VALIDATION accuracy over the T = 12 Euler grid, frozen, and only then evaluated once on test.

It helps in 18 of 36 conditions, mean +0.0018, best +0.0303 - a fix for the overshooting cases, not a free improvement everywhere. oracle_best, the best any t could have reached on test, is reported as an unreachable bound; the gap to it is .001-.005.

It rescues the only real regression

DTD / ResNet-18 / full, rolled-out — the worst cell in the project:

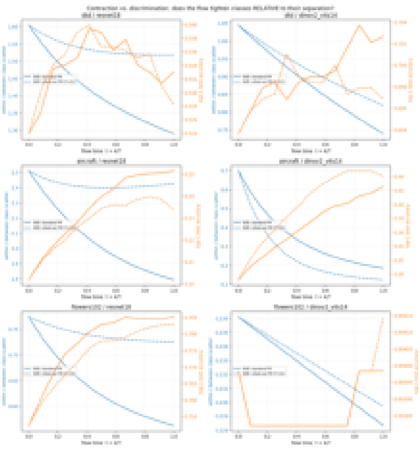
prototype baseline 0.5878
rolled-out FM at t=1 0.5672 (-0.0206)
rolled-out FM at t* 0.5975 (+0.0097)

Validation-selected early stopping turns a −.021 regression into a +.010 gain, at t* = 0.56. The strongest argument in the project for treating the stopping time as a hyperparameter rather than a convention.

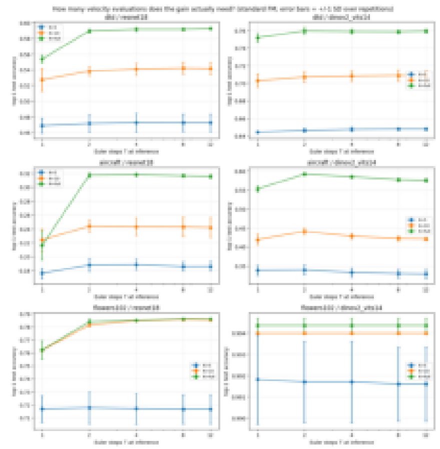
Reported as an ablation. The required Stage 2 table always classifies the final state z_T, exactly as ref/stage_2.pdf specifies. Rolled-out models overshoot hardest, which fits their weaker constraint. Not yet implemented on the CLIP branch, where it would be worth the most.

Contraction or discrimination? And how many steps?

Every metric on page 17 improves whenever the flow contracts toward the prototype set, whether or not the classes become easier to separate. Two ablations close that gap: the within/between class-scatter ratio W(t)/B(t), which falls only if classes tighten relative to their separation; and an inference-only sweep over the number of Euler steps.



W/B and accuracy over flow time



accuracy against Euler steps at inference

W/B at t=0 → t=1 (10-shot, standard FM)

setting	W/B	change	ΔAcc
FGVC-Aircraft · DINOv2	0.698 → 0.185	-0.5131	+0.1338
FGVC-Aircraft · ResNet-18	2.510 → 1.794	-0.7158	+0.0495
Flowers-102 · ResNet-18	0.774 → 0.562	-0.2116	+0.0330
DTD · ResNet-18	1.605 → 1.290	-0.3149	+0.0090
DTD · DINOv2	1.045 → 0.739	-0.3059	+0.0090
Flowers-102 · DINOv2	0.236 → 0.228	-0.0078	+0.0000

The step sweep is standard FM only — a rolled-out network cannot legitimately be run at a T it was not trained for — and the required table keeps T ∈ {4, 12}.

The flow discriminates, it does not merely contract

W/B falls in all 12 of 12 curves (median -0.219). But falling W/B is necessary, not sufficient: DTD sees a −0.31 improvement for under a point of accuracy. On Aircraft, cosine to the nearest competitor stays above cosine to the true prototype (.949 vs .942) — the negative margin again. Reorganising the cloud converts into accuracy only where the true prototype was not already ranked first.

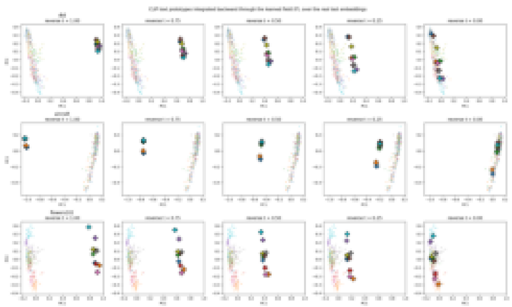
Two Euler steps are enough

T = 2 already delivers a median 100% of the T = 12 gain, and at least all of it in 11 of 18 settings. T = 1 delivers only 62% and is negative in three settings, where one large step overshoots to somewhere worse than the untouched feature. So the endpoint is T-independent for T ≥ 2: the layer costs two velocity evaluations, not twelve.

The flow in reverse — and what it reveals about CLIP

Start from the class prototypes at t = 1 and integrate the same learned field backward. The recovery rate asks whether a prototype, after that round trip, lands nearest to its own class's mean test feature. The untouched forward prototype is the pre-transport reference — a ceiling only on the image branch, where the prototype already is the class image mean.

branch · setting	forward	reverse	cos. distance moved
image · DTD / ResNet-18	0.957	0.957	0.0333
image · DTD / DINOv2	1.000	1.000	0.0261
image · FGVC-Aircraft / ResNet-18	0.660	0.530	0.0257
image · FGVC-Aircraft / DINOv2	0.690	0.620	0.0443
image · Flowers-102 / ResNet-18	1.000	1.000	0.0143
image · Flowers-102 / DINOv2	1.000	1.000	0.0055
CLIP · DTD	0.553	1.000	0.7175
CLIP · FGVC-Aircraft	0.250	0.750	0.6974
CLIP · Flowers-102	0.745	1.000	0.6772



CLIP text prototypes (P) integrated backward over the real test embeddings, at reverse t = 1 → 0. They start off to one side — that is the gap — and move in.

Image branch: almost nothing moves

Cosine distances of .006 to .044. The learned field is essentially zero near t = 1, so backward integration does not travel back toward the feature distribution. Recovery is preserved except on Aircraft, where it degrades. Well-behaved, but not usefully invertible — which is what a contraction should look like.

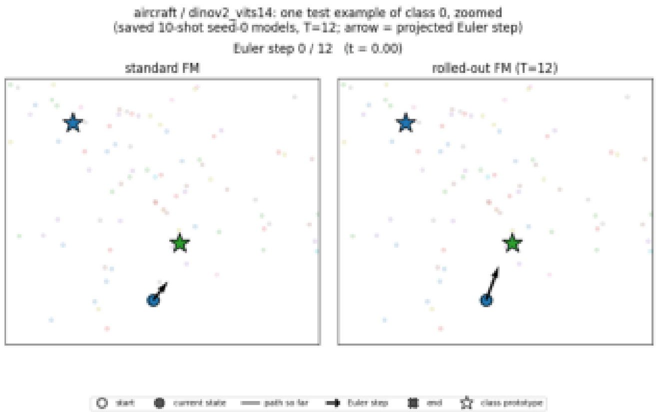
CLIP branch: the modality gap, measured

The prototypes move nearly orthogonally (≈.70) and recovery jumps from .55 / .25 / .75 to 1.00 / .75 / 1.00. The forward rate measures how far CLIP's text embeddings sit from its image embeddings; the reverse pass closes that distance. Direct evidence that the field learned to bridge the gap — and the reason the earlier "ceiling" wording had to be corrected: a text prototype does not start out inside its own class's image cloud.

Caveats unchanged: the backward pass evaluates the field at t ∈ {1, ..., 1/T} while the forward pass uses {0, ..., 1−1/T}, so this is not the exact inverse even for a perfectly learned field; and the forward flow is a deliberate contraction, so it is not invertible in principle. It recovers a plausible pre-image, not the original point.

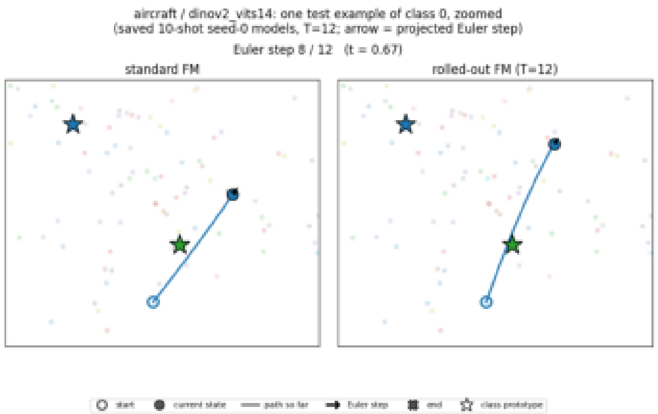
The transport, frame by frame

Four frames from the animation the notebook exports as a GIF: standard FM (left panel) and rolled-out FM (right) starting from an identical test feature, with the arrow showing the actual Euler update at that step. The setting is chosen programmatically as the largest-gain one — Aircraft / DINOv2, where the layer is worth +24.3 points.



Euler step 0 / 12 (t = 0.00)

Euler step 4 / 12 (t = 0.33)



Euler step 8 / 12 (t = 0.67)

Euler step 12 / 12 (t = 1.00)

Read qualitatively: the transport happens in DINOv2's full 384-dimensional space and this is a 2-D projection of it, so projected path length and curvature are not high-dimensional measurements. What the frames convey is the mechanism — T small updates accumulating into the representation the cosine rule finally classifies.

Findings

1 **Encoder choice dominates every method difference measured here.**

DINOv2 over ResNet-18 is worth 14–31 points. No baseline-versus-baseline gap in this project comes close.

2 **The FM layer works — but on fewer cells than the raw table suggests.**

It improves the prototype baseline in 66 of 72 cells, up to +24.3 points on Aircraft/DINOv2 at full data. Under a paired 95% CI only 39 of 72 are distinguishable from zero: 24/24 on Aircraft, 7/24 on DTD. The regressions are not significant either.

3 **Most of the largest gain is not specific to flow matching.**

FM beats a plain MLP and a time-free residual stack in 16 of 18 settings, but on Aircraft/DINOv2 the plain MLP already delivers 97% of FM's gain over the baseline. Flow matching earns its keep where that MLP overfits below the baseline instead.

4 **It repairs a broken classifier rather than sharpening a working one.**

Cosine to the true prototype rises everywhere, but margin rises only where the baseline was mis-ranked. In every setting, fixed samples started with negative margin and broken samples with positive margin.

5 **It does not beat a linear probe on the same features.**

Where the probe leads it recovers a median 59% of the gap (10–87%) and overtakes in only 4 of 18 settings, all near-saturated.

6 **Standard FM ≥ rolled-out training (14/18 settings), on accuracy and on compute.**

Rolled-out is the only variant that ever regresses below the baseline, the one that overshoots hardest, and the one that improves W/B least.

7 **t = 1 is a convention, not an optimum — but selecting it honestly is worth less.**

Accuracy peaks earlier in 6/6 CLIP curves and 6/12 image curves, up to +4.2 points. Validation-selected t* helps in half the conditions, and rescues the project's only real regression.

8 **Two Euler steps deliver the whole gain.**

Median 100% of the T=12 gain at T=2 against 62% at T=1. The layer is cheaper than the specified T suggests, and the T-independence has a floor at T = 2.

9 **On the CLIP branch, the fair control reverses the headline.**

FM beats zero-shot in 32/36 cells but loses to a same-supervision control in 28/36. Those gains measure the labels.

10 **The reverse flow makes CLIP's modality gap measurable.**

Backward integration lifts text-prototype recovery from .55/.25/.75 to 1.00/.75/1.00; on the image branch it barely moves anything.

Limitations and next steps

Limitations

- **n = 3 runs per setting.**

  The paired 95% intervals on page 8 are wide because of it, and that is the honest reading: 33 of 72 cells cannot be called either way.

- **Flowers-102 at K = 10 is one effective run.**

  The official train split is exactly 10 images per class, so all three subset seeds select the same images. Its std is 0 by construction, which also makes 9 of the 48 CI-significant cells degenerate.

- **The Stage 1 full prototype baseline is a single run.**

  It is closed-form, so it carries no error bar and the full-data deltas are unpaired.

- **No hyperparameter search.**

  The Stage 1 recipe was adopted as-is, per the specification. Better FM numbers are plausibly reachable with validation-only tuning.

- **2-D projections are qualitative.**

  The joint PCA plane explains only part of the variance; geometry off-plane is invisible. They are not evidence of classifier quality.

- **The CLIP branch has no paired CI and no t* selection.**

  Its "loses to the control in 28 of 36 cells" is still a difference of means, and it is the branch where early stopping would be worth the most.

- **One reporting column is unpopulated.**

  test_accuracy_sel_T needs one pass with FORCE_RETRAIN_STANDARD = True. The headline test_accuracy column is unaffected.

Next steps, in order of expected value

- **Isolate the time conditioning.**

  The controls showed that iterative shared-weight computation carries most of FM's advantage over a plain MLP. A residual stack with a t input and nothing else added would say precisely what flow matching contributes beyond that. This is now the most informative single experiment left.

- **Carry the paired CI and t* selection to the CLIP branch.**

  The image branch's headline changed materially once both were computed. Until they are run on the CLIP branch, its conclusions rest on differences of means.

- **Make the stopping time a hyperparameter everywhere.**

  It is selectable on validation data at no cost, and it converted the project's worst cell from −.021 to +.010.

- **Use the margin, not accuracy, to predict where the layer will help.**

  A negative baseline margin marks a mis-ranked classifier, which is precisely where prototype-targeted transport pays. A comfortably positive margin predicts the near-zero gains seen on DTD and Flowers-102.

> Every number in this report is parsed from the executed notebooks by doc/build_report.py; the diagrams on pages 2–3 are drawn for this document, and every scatter, trajectory, curve and animation figure is a real run output. Section 5b of 04_flow_matching.ipynb verifies that all 42 (dataset, encoder, K, seed) combinations reuse Stage 1's exact K-shot subsets, to a maximum prototype-reconstruction drift of 0.00e+00.